

Oozing Hand Crafted 2D Illustration Into a 3D Ninja Turtle World

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ACM Reference Format:

Jacques Daigle, Brian Gossart, and Olivier Mitonneau. 2024. Oozing Hand Crafted 2D Illustration Into a 3D Ninja Turtle World. In *Proceedings of SIGGRAPH Production Sessions '24*. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3641232.3649243>

1 Description

TMNT: Mutant Mayhem took on the monumental challenge of modernizing a very well-known and beloved franchise. It was important to the director, Jeff Rowe, that the audience believe the turtles were real teenagers... from their awkward bodies to how they acted (including their voices which were recordings of actual teenagers). The modeling also had to reveal all the imperfections related to their mutant nature, serving the idea of a teenager's insecure life. In line with this teenager vibe, the director and the production designer, Yashar Kassai, established the idea that the film design would suggest that it was created by an immature teenager drawing this movie in his sketchbook, with all of his naivety and energy.

The visual standards were very ambitious in that Mikros Animation was expected to deliver the quality and look of 2D concept art at each frame. To do that, Mikros and Paramount had to reconsider many parts of the classical CG workflow and pipeline, from asset creation to animation, through compositing.

There have been many explorations of bringing 2D aesthetics into 3D. Mutant Mayhem was unique in pushing this stylization at every step for a look that was not simply toon-shaded, but a sophisticated use of color and moods.

That includes: - The ability for modeling to simulate pencil lines with curves when developing their assets, giving another approach in the treatment of details. - Stepped animation without classical motion and the creation of pencil lines on the fly in Maya for various 2D motion blur effects and enhancements to expression. - A sophisticated lookdev workflow from surfacing to lighting, providing a stylized beauty pass to compositing with a good amount of control and graphic features. This included dynamic pencil grain applied on curve objects and the ability to give a painterly look to each light individually. The texture of the characters is not just painted, the lights themselves look like they are hand drawn. - Development in the lookdev of hair, one of the most challenging parts of giving a 2D look to a CG asset. Mikros approached the fabrication with procedural hair and a more specific shape to create a cohesive look.



- Custom tools in compositing to develop a show-specific defocus, motion blur, and flares throughout all sequences.

Building the stylization at each step gave the director a strong vision for the graphic quality of the assets and lighting out of render before the finalization of shots in compositing. All of these techniques were developed to give the best artistic leverage to each department, to add artistic and show specific features... and ultimately to create a handcrafted feel.

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SIGGRAPH Production Sessions '24, July 27 - August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0514-4/24/07

<https://doi.org/10.1145/3641232.3649243>